

Lecture 5: Numerical Methods And Linear Algebra

Essential Mathematics for Quants

Lecturer: Henry Chan

This lectures cover basic mathematical computation techniques commonly used in quantitative finance. It is essential for solving different numerical problems from day to day. Students are assumed to have basic understanding of calculus and linear algebra.

Please do not hesitate to reach me via WhatsApp, either in the group or PM, if you have any questions regarding the course.

Contents

1	Numerical Differentiation and Integration	1
1.1	Integration: Motivation	1
1.2	Integration: Numerical Approximation	2
1.3	Integration: Accuracy	2
2	Numerical Differentiation	3
2.1	Numerical Differentiation: Motivation	3
2.2	Numerical Differentiation: Accuracy	4
3	Linear and Nonlinear Equations	4
3.1	Linear Equations: Motivation	4
3.2	Linear Equations: Methods	5
3.3	Nonlinear Equation: Relaxation Method	6
3.4	Nonlinear Equation: Binary Search	6
3.5	Nonlinear Equation: Newton's Method	7

1 Numerical Differentiation and Integration

1.1 Integration: Motivation

You should have encountered different integration problem in your university studies which could be solved using closed form solution like the following:

$$\int_a^b x^2 dx = \frac{b^3}{3} - \frac{a^3}{3} \quad (1)$$

In particular, the university courses have taught different fancy techniques in getting the exact form of the integration. However, what if we encounter some crazy integration that could potentially happen in the real world?

$$\int_a^b \cos(x) e^{\sin(x)^2} dx = ? \quad (2)$$

Another question: Do we always need an exact solution?

In reality, it is not meaningful to have an exact solution. For instance, in finance, there no reason to compute something's value to a precision of 0.0000001 USD as there is always a minimal unit of notional that two parties could exchange.

This arises a more important question: how we can get a "good enough" calculation?

1.2 Integration: Numerical Approximation

Let's look an the integral

$$\int_a^b f(x) dx \quad (3)$$

The interval $[a, b]$ can be breakdown into sub-intervals $[x_{j-1}, x_j]$ with $x_j = jh$, $h = \frac{1}{N}$, and $a = x_0 < x_1 < \dots < x_N = b$.

The integral now can rewrite into

$$\int_a^b f(x) dx = \sum_{j=1}^N \int_{x_{j-1}}^{x_j} f(x) dx \quad (4)$$

When the N is large enough, the following term can be approximate as:

$$\int_{x_{j-1}}^{x_j} f(x) dx \approx h f(x_{j-1}) \quad (5)$$

Therefore,

$$\int_a^b f(x) dx = h \sum_{j=1}^N f(x_{j-1}) \quad (6)$$

1.3 Integration: Accuracy

There are different ways to do the approximation, and different approaches will offer a accuracy. To get the accuracy of the approximation, we have to know about Taylor series of a smooth function:

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(a) (x - a)^n \quad (7)$$

As long as the function $f(x)$ is N times differentiable, the Taylor series can be expressed as

$$f(x) = \sum_{n=0}^{N-1} \frac{1}{n!} f^{(n)}(a)(x-a)^n + \frac{1}{N!} f^{(N)}(y)(x-a)^N \quad (8)$$

Let's work on the simplest case where we take the Taylor expansion of $f(x)$ near $x = x_{j-1}$ for the integration $\int_{x_{j-1}}^{x_j} f(x)dx$.

$$\int_{x_{j-1}}^{x_j} f(x)dx = hf(x_{j-1}) + h \int_{x_{j-1}}^{x_j} f'(y(x))dx \quad (9)$$

We may write down the inequity,

$$h \left| \int_{x_{j-1}}^{x_j} f'(y(x))dx \right| \leq \max_{y \in [x_{j-1}, x_j]} |f'(y)| \int_{x_{j-1}}^{x_j} 1dx = h^2 \max_{y \in [x_{j-1}, x_j]} |f'(y)| \quad (10)$$

Therefore,

$$\int_{x_{j-1}}^{x_j} f(x)dx = hf(x_{j-1}) + O(h^2) \quad (11)$$

If we divide the range from $[0,1]$ into N pieces, each piece has a length of $h = \frac{1}{N}$. Then,

$$\int_0^1 f(x)dx \approx h \sum_{j=1}^N f(x_{j-1}) + O(h) \quad (12)$$

Instead of doing the Taylor expansion around x_{j-1} , the expansion could be done in the middle of the piece (i.e. $x_{j-1/2} = (j - 1/2)h$) instead of the leftest, which is called the midpoint method. After doing some math, it is

$$\int_0^1 f(x)dx \approx h \sum_{j=1}^N f(x_{j-1/2}) + O(h^2) \quad (13)$$

provided the function f is twice differentiable.

2 Numerical Differentiation

2.1 Numerical Differentiation: Motivation

Similar to integration, the objective is to compute a differentiation numerical to a sufficiently accurate level. Here are three common numerical method of doing differentiation.

$$\begin{aligned} \text{(Forward Difference)} \quad f'(x) &= \frac{f(x+h) - f(x)}{h} \\ \text{(Backward Difference)} \quad f'(x) &= \frac{f(x) - f(x-h)}{h} \\ \text{(Centered Difference)} \quad f'(x) &= \frac{f(x+h) - f(x-h)}{2h} \end{aligned} \quad (14)$$

2.2 Numerical Differentiation: Accuracy

Similar to integration, the accuracy can be estimated using Taylor expansion. For forward difference, Taylor expansion gives the following:

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(\epsilon), \quad \epsilon \in [x, x+h] \quad (15)$$

Therefore,

$$\frac{f(x+h) - f(x)}{h} = f'(x) + \frac{h}{2}f''(\epsilon) = f'(x) + O(h) \quad (16)$$

For centered difference, the error can be further reduced:

$$f(x \pm h) = f(x) \pm hf'(x) + \frac{h^2}{2}f''(x) \pm \frac{h^3}{6}f'''(\epsilon) \quad (17)$$

After some arithmetic, you will get that

$$\frac{f(x+h) - f(x-h)}{2h} = f'(x) + \frac{h^2}{6}f'''(\epsilon) = f'(x) + O(h^2) \quad (18)$$

This technique can be also applied to calculating the second derivative. For instance

$$\text{(Second Derivative)} \quad f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} \quad (19)$$

3 Linear and Nonlinear Equations

3.1 Linear Equations: Motivation

In the last lecture, we have discussed about dimensional analysis. Consider the simplest case:

$$\text{(Speed)} \ v = \frac{\text{(Distance)} \ d}{\text{(time)} \ t} \quad (20)$$

each has a dimension of:

- (Speed) $[v] : \frac{m}{s^{-1}}$
- (Distance) $[d] : m$
- (time) $t[t] : s$

For this example, it is easy to deduce (20) since the variables are simple. What if we are trying to deduce something more complicated from dimensional analysis? Let's say, market impact model solely using dimensional analysis.

$$P = c \cdot \sigma \cdot \sqrt{\frac{n}{v}} \quad (21)$$

with,

- P: Market Impact
- c: Constant
- σ : Volatility
- n: Order size
- v: Traded volume

There are 5 variables, with complicated dimensions. Without giving you the final form of the equation, it is rather hard to deduce the form from the dimension of variables provided. With linear equation, we may structure the problem nicely and use a well study tool to solve those problem. Let's return to the simple case (20) and structure it as a linear equation problem. We would like to write down the functional form of v using variable d and t , i.e.

$$v = f(d, t) = d^m \times t^n \quad (22)$$

The problem is equivalent to find m and n such that, the dimension of both side of the equation equals.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} m \\ n \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (23)$$

It can be written as a linear equation $Ax = b$! To solve the m and n in this case, or some even more complicated cases, there are standard methods that one may apply.

3.2 Linear Equations: Methods

In any linear algebra courses, they should have covered few major methods in solving the linear equation $Ax = b$, namely,

- Gaussian elimination
- Backsubstitution
- Pivoting
- LU decomposition

Regardless of the methods chosen, the ultimate goal is solving x in the equation $Ax = b$ by getting the inverse of A s.t. $x = A^{-1}b$.

We will not be covering the maths behind these methods since they are readily available online. Yet, most of the online materials didn't mention one thing: Accuracy and speed of different methods. When you are learning these methods, think about two things: 1) Time complexity, and 2) Error (especially for approximation algorithm).

3.3 Nonlinear Equation: Relaxation Method

What if you are solving equation like this

$$x = 2 - e^{-x} \quad (24)$$

There is no analytic method for solving this equation. We must rely on approximation or computational methods to solve it. Let us introduce the **relaxation method**. (We will jump into the coding to illustrate this method.)

The equation is first converted to $x = f(x)$ in order to apply the relaxation method. Whether the problem is solvable often depends on the starting value. Occasionally, rearranging the equation.

For example, $x = e^{1-x^2}$ is not solvable with relaxation method. After rearranging the terms to $x = \sqrt{1 - \log x}$, the equation becomes solvable. Why?

Assume x' is the value of x after an iteration by plugging in x .

$$x' = f(x) \approx f(x^*) + (x - x^*)f'(x^*) + \dots \quad (25)$$

By definition, $x^* = f(x^*)$, therefore,

$$x' - x^* = (x - x^*)f'(x^*) + O((x - x^*)^2) \quad (26)$$

After n iterations,

$$x_n - x^* = (f'(x^*))^n(x - x^*) + O((x - x^*)^2) \quad (27)$$

If $|f'(x^*)| < 1$, then,

$$x_n - x^* = e^{-n|\log(f'(x^*))|}(x_0 - x^*) \rightarrow 0 \quad (28)$$

In the example, as the $x^* = 1$, $|f'(x^*)| = |-2xe^{(1-x^2)}]_{x=1}| = 2 > 1$, which does not fulfill the requirement for relaxation method's convergence.

3.4 Nonlinear Equation: Binary Search

Binary search is a common method for solving nonlinear equations. In the following discussion, the non-linear equation is assumed to be arranged to become a form of $f(x) = 0$.

Algorithm 1 Binary search algorithm

```
1: Find an initial pair of  $x_1$  and  $x_2$  s.t.  $f(x_1)$  and  $f(x_2)$  have different signs.
2: Compute the midpoint between  $x_1$  and  $x_2$ :  $x' = \frac{1}{2}(x_1 + x_2)$ .
3: if  $f(x') = 0$  then
4:   Root is found.
5: else if  $f(x')$  has the same sign as  $f(x_1)$  then
6:   Set  $x_1 = x'$ 
7: else
8:   Set  $x_2 = x'$ 
9: end if
10: if  $|x_1 - x_2| \geq \epsilon$  where  $\epsilon$  is the target accuracy then
11:   Repeat the loop
12: else
13:   Take  $x^* = \frac{1}{2}(x_1 + x_2)$  as the final solution.
14: end if
```

3.5 Nonlinear Equation: Newton's Method

Newton method is one of the most common root finding method that utilizes the first derivative (i.e. slope) of the equation to find the root.

Slope of a function $f(x)$ at point x can be written as

$$f'(x) = \frac{f(x)}{\Delta x} \quad (29)$$

In each update, the updated x' would be

$$x' = x - \Delta x = x - \frac{f(x)}{f'(x)} \quad (30)$$

To estimate the rate of convergence of newton's method, let's consider the Taylor expansion of an arbitrary estimation x of the root, and x^* is the true root of the function.

$$f(x^*) = f(x) + (x^* - x)f'(x) + \frac{1}{2}(x^* - x)^2 + \dots = 0 \quad (31)$$

After rearranging the equation,

$$x^* = \left(x - \frac{f(x)}{f'(x)}\right) - \frac{1}{2}(x^* - x)^2 \frac{f''(x)}{f'(x)} + \dots = x' - \frac{1}{2}(x^* - x)^2 \frac{f''(x)}{f'(x)} + \dots \quad (32)$$

For the error of the estimate x , it will be $x^* = x + \epsilon$. For the next estimation, it's error will become $x^* = x' + \epsilon'$. Therefore,

$$\epsilon' = \left(-\frac{f''(x)}{f'(x)}\right)\epsilon^2 \quad (33)$$